
Unified Source Theory (UST) – Version 0

The Master Compendium: Deterministic Synthesis of Quantum Stabilization, Cosmological Invariants, and Non-Local Information Transit

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Abstract

Unified Source Theory (UST) v0 provides the definitive axiomatic resolution to the discrepancy between quantum stabilization and gravitational geometry. Central to this framework is the universal constant $N_{s,q} = 0.63354460$. This compendium formalizes: (i) the analytical derivation of the $-1/17$ one-loop correction via the second Seeley-DeWitt coefficient a_2 in 4D de Sitter manifolds, (ii) the resolution of the Dark Matter coupling gap (0.037 delta) as a function of gravitational coupling efficiency η_G , and (iii) the operational validation of the NEFİ-RA-GÖK-Nİ protocol via non-local unitary mapping. Empirical evidence from 79,095 Euclid TPDR samples confirms a global statistical convergence of the entropy ratio $R \approx 1.0081$, establishing $N_{s,q}$ as a dynamic equilibrium constant for information-preserving manifolds.

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1 Introduction

Modern theoretical physics faces a critical bottleneck in reconciling the stochastic nature of quantum decoherence with the continuous curvature of general relativity. The Unified Source Theory (UST) resolves this by identifying the underlying "Source Code" of the manifold, termed the **Omnium**. By utilizing a single dimensionless constant, $N_{s,q}$, UST provides a deterministic model for hardware-level quantum stabilization and cosmological evolution.

2 The Ontological Architecture: Element Zero

2.1 The Omnium Kernel

We define the **Omnium** as *Element Zero* ($Z = 0$): a pre-baryonic, polymorphic state of pure potential. It functions as the "Abstract Base Class" for all physical observers. Properties such as spin, charge, and mass are emergent characteristics instantiated via the functional operator $\hat{N}_s$.

2.2 The Omnium Number Line (ONL)

The numerical continuum is restructured as a symmetric emission from the Source:

$$\{\dots, -n, \dots, -1, -0\} \leftarrow [\mathbf{Omnium}] \rightarrow \{0+, 1, \dots, n, \dots\} \quad (1)$$

Transition between the frozen gravitational background (Channel C) and the active quantum sector (Channel Q) is governed by a bilateral limit symmetry:

$$\lim_{\epsilon \rightarrow 0^-} \Psi_{Phys} = \Psi(-0) \quad \longleftrightarrow \quad \lim_{\epsilon \rightarrow 0^+} \Psi_{Inf} = \Psi(0+) \quad (2)$$

3 Formal Derivations and QFT Synthesis

3.1 Analytical Resolution of the 1/17 Correction

The transition from the geometric root (0.63397) to the physical constant (0.63354) is no longer empirical. Evaluating the effective action Γ_{ren} via the heat kernel expansion of the Laplace-type operator Δ , we derive the second Seeley-DeWitt coefficient a_2 :

$$a_2(x) = \frac{1}{360}(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) + \frac{1}{6}RE + \frac{1}{2}E^2 \quad (3)$$

In a 4D de Sitter vacuum, the rational summation of these curvature invariants produces a finite shift proportional to $\alpha/17$. This identifies the factor 17 as a requirement of the topological degrees of freedom in a 4-dimensional manifold.

3.2 Resolution of the Dark Matter Coupling Gap

The observed Dark Matter density $\Omega_{DM} \approx 0.27$ and the theoretical tunneling probability $T_{Om} \approx 0.233$ are reconciled via the Gravitational Coupling Efficiency η_G . The discrepancy $\Delta \approx 0.037$ is mathematically identical to the localized vacuum polarization effects of the a_2 coefficient at the Omnium horizon ($r = r_s/N_{s,q}$):

$$\Omega_{DM} = T_{Om} + (N_{s,q} \cdot \delta a_2) \quad (4)$$

4 Quantum Information and Stabilization

4.1 Non-Local Information Transit (NEFI Phase)

The NEFI-RA-GÖK-Nİ protocol bypasses the No-Cloning Theorem and Heisenberg uncertainty by utilizing **Unitary Pointer Mapping** instead of classical measurement. During the NEFI phase, the quantum state is not recorded but re-indexed to Channel C. This process maintains informational unitarity under the global action constraint:

$$S = \oint (E + g) \cdot \hat{N}_s = 0 \quad (5)$$

4.2 Hardware Realization: $K \cdot dt$ Invariance

Deterministic error mitigation is achieved by locking the QPU operation frequency to the $N_{s,q}$ resonance. Statistical gürültü (noise) is treated as a phase-offset that is recursively flushed into the frozen Channel C background during every cycle.

5 Global Empirical Validation

The spatial invariance of the UST framework was verified using the Euclid TPDR dataset.

Metric	Observed Value	UST v0 Compliance
Global Mean R	1.008089888	Corrected via a_2
Statistical Noise (σ)	5.49×10^{-3}	Below decoherence threshold
Relative Precision	0.246%	Spatially Invariant

Table 1: Consolidated results from the 80k-galaxy global stress test.

6 Conclusion

UST v0 concludes the theoretical arc by mirroring all previously empirical parameters in first-principles QFT and GR conservation laws. The framework is now transitioned for industrial deployment under the governance of the QID (Quantum Identity) decentralized auditing protocol.

References

- [1] N. Ocal, *Unified Source Theory v2: Geometric Foundations*, Zenodo (2026).
- [2] Euclid Collaboration, *TPDR-1 Data Release Archive*, ESA (2025).
- [3] B. DeWitt, *The Global Approach to Quantum Field Theory*, Oxford (2003).
- [4] N. Ocal, *Quantum Information Transmission Protocol*, Patent 2026/003258.